PLECTA: Overlap-aware instance reconstruction of dense filament networks from binary masks

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Abstract

Dense filament networks resist instance segmentation because their two failure modes pull in opposite directions: crossings join distinct objects into one connected component, while gaps in the segmentation divide a single object into several. We present PLECTA, which converts a binary filament-axis mask into overlap-aware two-dimensional instances. It decomposes the skeletonised mask into arms and junctions, alternates exact per-junction continuation matching with globally gated gap matching, and re-estimates tangent, chord, and curvature evidence from the assembled chains, so that a crossing pixel may belong to more than one output instance. What is new is narrow and specific: each junction of any degree is solved as one exact maximum-weight *partial* matching in which leaving an arm unmatched is priced rather than left over, so a filament can end instead of having a continuation invented for it. What makes that decidable is directional continuity: the evidence is entirely directional, tangent and curvature read over tens of pixels of arclength either side of the discontinuity, so the method applies where direction persists across a crossing, a condition bending stiffness contributes to and rod-like filaments commonly meet. That is the semiflexible regime occupied by nanotube bundles, cytoskeletal filaments and synthetic fibres, and it excludes freely flexible chains, where no local direction survives the junction to be matched. Skeleton graphs, continuation angles, distance gates, optimisation and overlapping output are prior art. The method draws on no random number generator and its configuration is published, so a mask yields the same instances on every run.

Evaluation used a method-independent partition of each input skeleton into common fragments, and reports pairwise F_1 with the adjusted Rand index and variation of information beside it. On 128 independently seeded held-out scenes from the same procedural generator, PLECTA achieved mean pairwise F_1=0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]), precision 0.868, recall 0.843 and adjusted Rand index 0.851, with no failed scenes, falling from 0.911 at 20% coverage to 0.775 at 60% and identifying dense crossings as the main remaining regime of error. Against a greedy minimum-turning-angle continuation rule on identical masks it gained +0.145 F_1 [+0.129, +0.162] over 50 scenes, and +0.478 and +0.339 over two released implementations run from their own sources; it leads at every coverage on this generator under all three reported measures, which establishes a ranking on this material and not a general one. Of the 3638 reference crossings it partitions the incident arms exactly as the reference does at 0.809 of them, against 0.088 for leaving every arm unpaired.

CNT-sheet scanning electron microscopy is the proof-of-concept application: an upstream synthetic-data-driven CycleGAN/U-Net route prepares the binary masks that let the method run on real micrographs, while image-informed width measurement and ribbon rendering are downstream stages that leave the grouping fixed. The evidence supports same-generator synthetic generalisation and a demonstration on 3 annotated SEM fields, two of them held out from the upstream segmenter, and not general real-image or three-dimensional reconstruction claims.

Keywords: filament networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

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[Igor 1] I must say I'm not quite happy with FilaSeg - we need to come up with a better name of the software. I think we do more than just "Seg", I'd avoid that in the name anyways. I would not go towards something trivial like FIBER or STRING either - not enough distinctive. I find attractive something like REUSS, OCCAM or COSSERAT - with reference to physics/mechanics besides the meaning in abbreviation. But that's your baby of course, your word will be the last.

[Response] Addressed. The method is now PLECTA, after Latin *plecta*, a braid or twisted rope. It carries a meaning tied to what the method does, avoids *Seg*, and is not generic. No expansion is forced onto the letters.

1 Introduction

[Igor 2] The paper definitely should have one (and only one) focused, bright central idea. My preference - we present a black box that magically transforms microphotographs of entangled bundles of sticky elastic rods into high-resolution, segmented 3D models. We can frame it as the story about CNTs and make it carbon-specific or talk about arbitrary systems (spagetti, straw, bone tissue etc) and make it an image analysis paper. But we certainly should not pick separate elements of the pipeline and discus only those - this way the value of the paper becomes incrementally small.

[Response] Open decision, not yet settled. The single central idea here is *recovering individual filament identity through crossings*, and the paper is built around it end to end: micrograph, mask, instances, geometry. Two constraints pull against the wider framing. The reconstruction is projected and two-dimensional, so *3D models* would overclaim unless we add depth evidence; and the upstream segmenter is a separate contribution with its own evaluation, so folding it in would mean defending two methods at once. The current draft therefore keeps the CNT story as the application and the identity problem as the idea. Please say if you would rather we widen it.

Separating the individual objects in an image of a dense filament network is an instance-segmentation problem with an awkward geometry. The objects are long, thin and open, they touch and cross and occlude one another, and what distinguishes them is local continuation rather than shape or extent. This paper presents a method for that problem. Its proof-of-concept application is scanning electron microscopy (SEM) of carbon-nanotube (CNT) sheets, whose mechanical and electrical response depends on the geometry of the bundle network: length, width, orientation, curvature and contact topology [1–5]. SEM resolves that projected network, but routine image analysis often stops at coverage or orientation distributions [6, 7], whereas instance-resolved analysis requires each visible bundle to remain addressable across gaps, touches and crossings.

That requirement is distinct from semantic segmentation. U-Net and its self-configuring variants can identify filament foreground [8, 9], but a binary foreground mask does not encode object identity: a faded segment splits one filament into several connected components, and a crossing joins different filaments into one. The ambiguity is greatest in dense networks, where several local continuations are geometrically plausible. Compact-object instance priors (bounded blobs, centre-directed flows, star-convex shapes) are poorly matched to long open curves that may overlap [10–12]. Ridge and vesselness filters enhance curvilinear signal but do not assign identity across junctions [13–15], and fibre extraction packages such as FIRE, CT-FIRE, SOAX, FiNTA and DiameterJ report network statistics rather than resolved instances [16–20].

Recovering individual filaments through crossings is itself established prior art, and the design space is well explored. Basu et al. build a minimum spanning tree over centreline particles, delete every bifurcation node, and recombine the segments with a binary integer program over chains of at most three, priced by the integrated squared curvature of a fitted spline; a segment may stay unlinked, and the program is re-solved after refitting until the segment set stops changing [21]. DeFiNe consumes a pre-extracted weighted graph and solves a filament cover as a linear program minimising path roughness, optionally letting several paths cover one edge, so overlaps are representable [22]. SIFNE excises each crossing with its neighbourhood, estimates tip directions towards the local centre of mass, and pairs termini greedily in priority order inside a fan-shaped region under angular and continuity gates, leftover tips becoming ends [23]. De et al. resolve crossovers by partitioning a directed graph [24], and Wegmayr et al. group skeleton branches by lowest mutual angle and learn pixel embeddings for tangled microplastic fibres, sharing junction pixels in principle but assigning them arbitrarily [25]. DNAi detects junctions morphologically, clusters them by single linkage, erases a disk the width of the fibre, and enumerates the pairings at junctions of degree three or four from endpoint angles traced over a short span [26]. GraFT builds a graph from a Frangi-filtered skeleton, derives an angle threshold per component, and traces filaments by constrained depth-first search and a greedy longest-first path cover [27]. Learned alternatives include orientation-aware terminus pairing [28], FibeR-CNN, and recurrent pick-and-trace models [29, 30].

Two of those methods are compared against directly here, so what each was built for matters, because neither was built for this. DNAi is an end-to-end tool for DNA fibre spreading assays, in which immunofluorescence images of replicating DNA carry two nucleoside-analogue labels and the quantity wanted is the length ratio of the two tracts; it matches trained human annotators on that task and is released with both its models and its annotated data [26]. GraFT traces and tracks filamentous structures through time in confocal series of the plant actin cytoskeleton, and the temporal dimension it handles is outside the scope of this work [27]. Both are strong in their own domains, and both make design choices those domains license: DNAi commits to a pairing at every junction of degree two to four and attempts none above that, while GraFT's greedy longest-first path cover cannot revise an early commitment. Such choices cost little where filaments are sparse and well separated, which is the regime each was validated in, and more where filaments are dense and crossings are frequent. Section 3.3 runs both from their own released sources, on identical input masks and through one scorer, and reports what that costs on this material.

Skeleton graphs, junction excision, continuation angles, distance gates, combinatorial optimisation, iterative refitting and overlapping filament output are therefore all prior art, and none is claimed here. What is new is narrower, and it is a combination rather than any single element: from a binary mask alone, PLECTA solves each junction of any degree as one exact maximum-weight *partial* matching in which leaving an arm unmatched is an explicit option at a fixed price rather than a parity leftover, alternates that junction matching with globally gated gap matching, re-fits its tangent and curvature frames along the chains as they assemble, and emits overlapping instances. Each element exists somewhere in the catalogue above; no method there combines them. The methods that optimise globally consume intensity evidence or a pre-extracted weighted graph rather than a mask, and Basu et al. recombine chains of at most three segments [21, 22]; the mask-reading methods pair endpoints greedily in priority order [23, 25, 27] or enumerate the pairings of a junction only up to degree four [26]. Degree-general exact matching with a priced refusal is what lets a filament genuinely end instead of having a continuation invented for it.

That claim carries a condition, and it is worth stating as scope rather than leaving to be inferred. The only evidence PLECTA has at a crossing is directional: the reversal angle between two outward tangents, the turns onto the chord between them, and the continuity of signed curvature, each estimated over the 24 to 55 px of arclength the frames are fitted on. The working condition is therefore a property of the image: tangent and curvature must persist across the junction at the scale of that support, or the continuation is not decidable from geometry at all. Bending stiffness is the material property that most directly produces such persistence, which is why the semiflexible regime, where the persistence length is long against the width of a crossing, is the method's home ground; carbon-nanotube bundles sit there, and so do the actin, collagen and synthetic-fibre networks the methods above were built for. But stiffness contributes to the condition rather than uniquely causing it: tension, adhesion, confinement or bundling can hold a projected direction just as straight, and a stiff filament can still be ambiguous at a symmetric crossing. Where the condition fails, as for a freely flexible chain whose tangent decorrelates within the junction, no continuation is more plausible than another, and no amount of matching recovers what the image no longer distinguishes. The closest single prior method is that of Liu et al., which likewise works from a mask, reconnects across gaps and permits overlapping output, but pairs termini greedily and depends on a trained orientation network [28].

We introduce PLECTA (Pairwise Linking with Evidence from Chain Tangents and Assignment; the name is also the Latin *plecta*, a braid), a graph-matching method for dense filament networks. PLECTA decomposes a skeleton into arms, endpoint stubs, and merged junction nodes. It then alternates exact per-node continuation matching with globally gated gap matching while tangent and curvature estimates expand from local arms to assembled chains. Its output is a set of projected 2-D instance layers in which crossing pixels may be shared. Every parameter is fixed in a published configuration and the implementation draws on no random number generator, so a given mask yields the same instances on every run; that is the only sense in which it is called deterministic here.

Reaching real micrographs takes one more component, and the closest complete workflow is that of Peters et al., who join a U-Net-like semantic segmentation, trained on more than a thousand annotated SEM micrographs, to algorithms that search, trace and refine fibre-shaped segments through intersections, and validate the result against human evaluators on fibre number, length and width over six CNT materials [31]. That work and this one divide the task at the same joint, semantic foreground first and instances second, and differ in what the instance step is asked to prove: their tracer is validated as a substitute for human counting and measurement, morphological statistics on materials of graded complexity, while PLECTA is isolated behind a binary-mask contract and scored on identity through crossings against overlap-aware references in which every pairing is known. The two are complementary rather than competing readings of the same architecture, and the grouping stage evaluated here could in principle serve behind their front end as behind ours. We divide the workflow in two: Pipeline A turns a raw SEM micrograph into a binary filament-axis mask by a synthetic-data-driven CycleGAN/U-Net route whose training pairs are synthesised rather than annotated [32, 33], and Pipeline B is PLECTA. Without Pipeline A the method could only ever be exercised on masks drawn by hand or emitted by a generator; with the division, grouping can be measured on its own against references whose identities are known exactly, and width rendering cannot move an identity score. Pipeline A carries its own evaluation [33] and is not re-evaluated here. We evaluate PLECTA on 128 unseen scenes from the development generator and on 50 paired clean and degraded scenes, and demonstrate the two pipelines together on 3 annotated CNT-sheet SEM fields, 2 of them held out from the upstream segmenter.

2 Materials and methods

2.1 Scope and workflow

The formal input is a binary filament-axis mask $M \in \{0,1\}^{H\times W}$, from which PLECTA reconstructs an overlap-aware collection $\{I_1, \ldots, I_N\}$ with $I_n \subseteq M$ and a crossing pixel possibly belonging to several instances. The evaluated grouping stage uses only M; a grayscale SEM image is optional downstream input for width and brightness measurement and cannot revise the fixed grouping.

Figure 1 places the method in the two-pipeline workflow of Section 1: Pipeline A produces the mask from a micrograph, and Pipeline B, which this paper evaluates, is PLECTA. Its five geometric steps read mask geometry alone, and the two matching steps are re-made over eight rounds against frames re-fitted along the chains assembled so far.

What makes the task hard is a single crossing: the mask holds one connected blob from which several arms leave, and nothing local states which arm continues into which. The output is a set

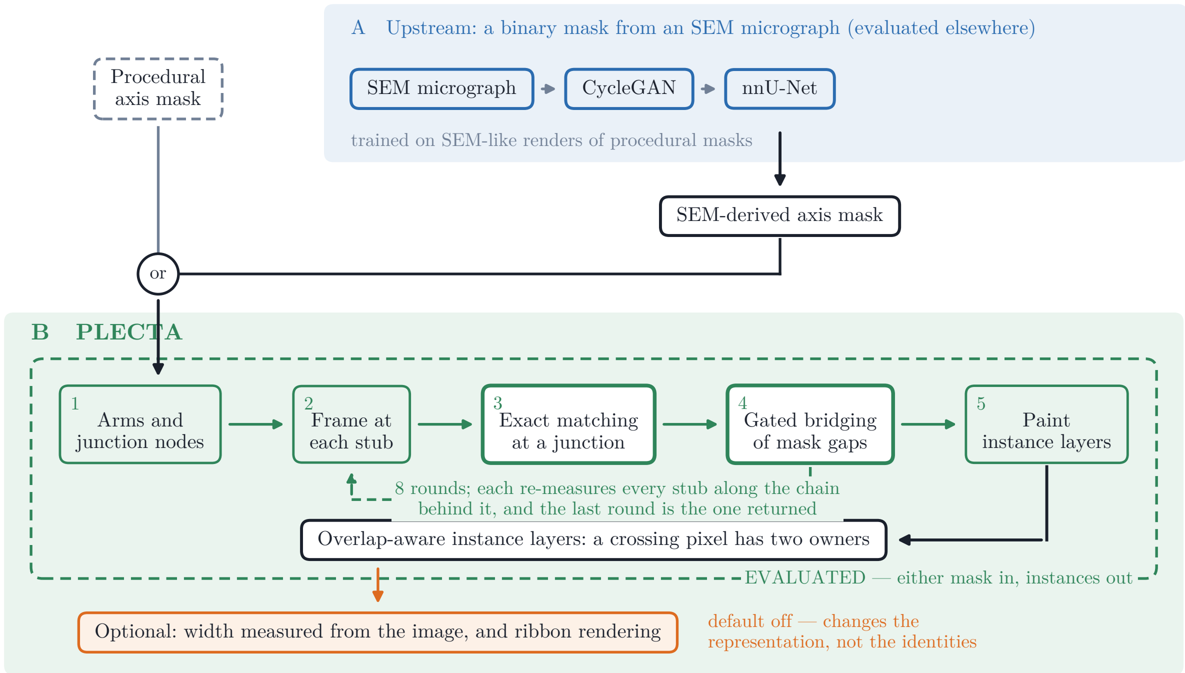

Figure 1. Where PLECTA sits. Pipeline A is the upstream binary segmentation of SEM micrographs, which carries its own evaluation; Pipeline B is PLECTA, and every result in this paper measures the dashed green box only. The mask entering Pipeline B came from Pipeline A for the SEM fields and from the procedural generator in every synthetic experiment. Steps 3 and 4 are the only decisions the method makes and are drawn heavier for it; everything else is bookkeeping. Colours: blue, a learned component; green, a PLECTA geometric step; orange, a dependency on the grayscale image; dark, a data artefact.

of layers rather than a partition, so a pixel inside that blob is counted in both instances that claim it.

2.2 Synthetic data and evaluation split

Procedural scenes preserve the ordered centreline $\mathcal{C}_k$ and rendered support Ω_k of every filament k; supports remain separate, so several filaments may own the same crossing pixel. Besides the exact overlap-aware reference, the generator provides an undegraded one-pixel axis mask, a degraded thin mask containing small gaps and irregularities, a combined thick foreground, and an SEM-like image. The degraded one-pixel axis is the input evaluated throughout. Areal coverage is the union area of the thick supports divided by image area, not the occupancy of the thin input mask, and Section 3.2 shows it is a proxy for difficulty rather than a cause of it.

Development used 84 scenes spanning 20–60% coverage. The held-out set comprised 128 different seeds, 32 scenes at each of 20, 30, 40, and 60% coverage: an independent sample from the same generator, not an independent distribution. A separate 125-scene locked set remains unevaluated. The compact seed manifest records every seed block and an executable verifier checks counts, disjointness, and available scene metadata.

A separate controlled paired set of 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage, was scored once from each geometry's clean axis and once from its degraded mask, after the configuration was fixed.

2.3 CNT SEM data and mask sources

3 manually annotated CNT-sheet SEM fields were available for exploratory analysis, each 1536 × 1024 px: B58_100 (307 visible CNT bundles), B58_110 (430) and B58_300 (325).

The annotated instance is the visible bundle, not the individual nanotube, which SEM does not resolve at this magnification. The nominal horizontal field width is 1.66 µm across the long, 1536-px axis of the stored images, which is 1.08 $\mathrm{nm\,px^{-1}}$. Each annotation is one human interpretation of projected, locally ambiguous crossings, and B58_100 and B58_110 informed development.

The reference annotations were drawn with an in-house interactive instance-segmentation tool for greyscale SEM of thin, crossing filaments, whose data model is per-instance masks rather than a flat label image, so a pixel may belong to several instances at once; that is what makes the reference overlap-aware and comparable with PLECTA's output at all. Its primary artefact, a compressed sparse per-instance pixel set, is the identical file the evaluation loads as its reference, so provenance is end to end. The annotator paints each visible bundle as its own instance, and the tool assists without deciding: strokes may snap onto the local intensity ridge, with polarity calibrated from the existing annotation, and may be re-rendered through a smoothing-spline fit whose centreline is shifted onto the measured intensity crest only where that crest is consistent and by at most about 8 px; widths come from perpendicular full-width-at-half-maximum profiles referenced to the local minima either side of the peak and reported as twice the nearer half-width, so a crossing filament fusing into the profile does not bias the measurement wide. Candidate gap bridges, collinear endpoint pairs of different instances, are suggested but never applied automatically; each is accepted or rejected by the annotator, so identity through every crossing is a human decision, machine-assisted but not machine-made. The tool keeps an audit log of every mutating action alongside the saved annotation, and an advisory per-instance plausibility score, computed from profile support, peak contrast and width plausibility, was used as quality control while annotating; the score is advisory only and is not part of the saved reference.

The protocol's limits are stated rather than implied. There was one annotator, who is an author of this paper and also configured PLECTA; there is no second reader, no blinding and no inter-reader agreement estimate, so differences smaller than plausible annotation variation should not be read as real, and the annotations are manual reference annotations, not ground truth. One assumption is also shared across the divide: the annotator's width measurement and PLECTA's optional width layer of Section 2.5 both read perpendicular FWHM profiles across a centreline. That does not touch the grouping scores, which are computed on centreline fragments, but it does mean the two are not independent about width; nothing in this paper compares the two widths, and no such comparison should be made without restating this.

Two inputs were derived per field. The first is the skeleton of the manual instance annotation, an oracle-like condition that asks what the grouping can do when the axis evidence is curated. The second is the output of Pipeline A, which asks what an automatic mask allows it to do. Pipeline A adapts the open CycleGAN/U-Net particle-segmentation workflow of Rühle et al. [32]: the procedural generator of Section 2.2 draws a binary mask, a CycleGAN translates it into an SEM-like image so that image and mask are paired without an annotator, and a 2-D nnU-Net [9] is trained on that paired corpus with the residual-encoder XL configuration and its heavy-augmentation trainer, all folds, best checkpoint. Its prediction is registered into the annotation grid and thresholded to give the axis mask PLECTA is handed. Pipeline A is a separate contribution with its own evaluation and its full configuration [33]; what matters here is only that its output is a binary mask produced without seeing any annotation.

Which fields that model had seen is a property of the (model, field) pair and was verified by image content rather than by filename. B58_110 and B58_300 are absent from the training set of the run used here; B58_100 is present in every available run and its U-Net condition is therefore an upper bound throughout. Section 3.7 measures what that contamination is worth rather than leaving it as a caveat. Mask quality was compared after skeletonisation with the symmetric 2-px tolerance used throughout, which avoids rewarding mask thickness, and reconstruction scores were computed within each mask's own visible common-fragment population.

[Needed before submission: Author metadata: SEM instrument and detector,

accelerating voltage, working distance, specimen preparation, and CNT material identity.]

2.4 PLECTA reconstruction

2.4.1 Skeleton graph and local frames

PLECTA binarises M at > 0, skeletonises it, and iteratively removes spurs no longer than 3 px. Skeleton pixels with degree at least three form junction clusters, and adjacent clusters are merged into nodes. Connectors up to 5 px are absorbed into one node; connectors up to 14 px merge nearby nodes but remain real arms; free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm with one stub at each end.

A stub frame contains its tip, outward tangent, and signed curvature, estimated by an arclength polynomial fit and considered reliable only with at least three samples spanning at least 3 px. Local frames use 24 px of support; once links form chains, the support expands to 55 px, so that a stub too short to have a direction of its own borrows one from the chain it has joined.

The cost of linking stubs a and b combines three angular terms, curvature continuity and, for gaps, a length penalty (Figure 2):

$$C_{ab} = w_d\,\theta + w_t\,q(d)\,(\varphi_a + \varphi_b) + w_\ell\,\frac{d}{\ell_0} + 30\,w_\kappa\,|\kappa_a + \kappa_b|, \qquad q(d) = \min\left(1, \tfrac{d}{d_0}\right), \tag{1}$$

with d the tip separation, θ the reversal angle between the two outward tangents, and φ_a, φ_b the turns from each tangent onto the tip-to-tip chord, measured at their own vertices. Junction links set $w_\ell = 0$, a junction having no gap to pay for. The reversal angle ignores the chord, which keeps it meaningful when the tips are three pixels apart and the chord direction is quantisation noise; the chord turns instead see lateral offset, so two parallel arms that never meet are penalised. Because it is the chord terms that degrade over short separations, the fade $q(d)$ withdraws them as the tips approach rather than letting an unstable angle dominate. The constant 30 places curvature, in px^{-1}, on the scale of the angular terms; the weights, ℓ_0 and d_0 are fixed in the public configuration.

2.4.2 Candidate gates and exact matching

Leaving a stub unmatched is priced rather than forbidden, and the price p_i also fixes admissibility: an edge is offered only when $C_{ij} < p_i + p_j$. Exact maximum-weight partial matching then maximises $\sum(p_i + p_j - C_{ij})$ over disjoint pairs, which minimises the total link cost plus p_i for every stub left free. At the configured prices junction stubs carry $p_i = 0.62$, so a junction edge needs $C < 1.24$; gap edges need $C < 0.60$ and must in addition satisfy $d \le 85$ px, $\theta \le 0.40$ rad and $\varphi \le 0.28$ rad at each end. Gap gates are the tighter set because a gap link asserts a continuation across evidence that is missing rather than merely ambiguous; the values themselves were fixed on development scenes, and the complete set is recorded in the public configuration. Rounds $r = 0, \dots, 7$ apply a schedule fraction $\alpha_r = 0.85 + 0.15\,r/7$ that scales the unmatched prices and the two gap angle limits and nothing else. Scaling a price *down* admits fewer edges, admissibility being $C_{ij} < p_i + p_j$, and scaling an angle limit down narrows the gap gates, so early rounds are the conservative ones, deliberately under-linked: while the frames are at their least reliable, only pairings that survive the reduced prices and narrowed gates are committed, and the constraints relax towards their configured values as the chains assembled so far sharpen the frames. The final round, $\alpha_7 = 1$, is the only one run at the configured parameters and is the one returned. In full:

1. Skeletonise M, prune spurs, merge junction clusters into nodes, and store the remaining paths as arms with a stub at each end.

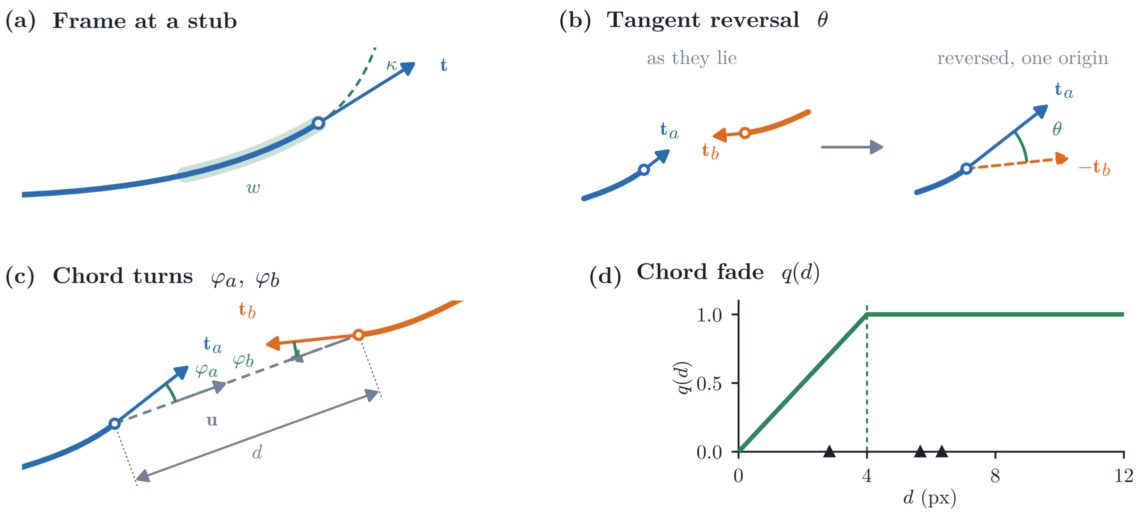

Figure 2. The stub frame and the terms of the pairing cost. (a) The least-squares arclength fit over window w, read outward from the tip at $s = 0$, giving outward tangent $\mathbf{t}$ and signed curvature κ; the fit is quadratic when the span is at least 18 px and linear otherwise. (b) $\theta = \arccos(\mathbf{t}_a \cdot -\mathbf{t}_b)$, in the two steps its definition takes: the two outward tangents where they lie, then the same pair with $\mathbf{t}_b$ reversed and both read from one origin, which is where the angle is measured. A perfect continuation makes the two tangents antiparallel, so $\theta = 0$. (c) The chord turns φ_a, φ_b onto the chord $\mathbf{u}$ joining the tips. (d) The fade $q(d)$ against tip separation; triangles mark the six real separations at the crossing of Figure 3.

2. For $r = 0, \ldots, 7$, re-solving both matchings from scratch and carrying only the chain structure forward from round $r - 1$:
   (a) Re-fit every stub frame along the chain it now belongs to.
   (b) At each node, form candidate edges between stubs of different arms with finite frames and cost, keep those admissible at α_r, and solve one maximum-weight matching per node.
   (c) From the still-unmatched reliable stubs, form gap candidates on different arms and different nodes that pass the distance gate and the α_r-scaled angle gates, and solve one global maximum-weight matching over the admissible ones.
3. Return the links of round $r = 7$.

Instances are assumed to be open projected curves, and a guard enforces it: after each of the two matchings, any matched component whose links close a ring has its longest link removed. On the 84 development scenes the guard never fired, zero links removed, so it protects against a configuration the data did not produce rather than shaping the result; closed or looping filaments are outside the scope of this paper.

Every stub is therefore either paired or pays the unmatched price, and there is no third possibility. The priced option is what lets a filament genuinely end, and what stops a T-junction from inventing a continuation for the arm that has no partner. The admissibility limit is not a separate threshold: an edge with $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching would never select it, and refusing to offer it removes an edge that could not have been chosen.

Solving a junction as one matching rather than one edge at a time is the distinction claimed in Section 1, and Figure 3 shows what it buys on real nodes, alongside the other two decisions the linker makes. A locally attractive edge can be declined, as at the three-arm node of Figure 3c. Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense: over 20 development scenes they never disagreed at nodes of degree 2–4 ($n = 542$), disagreed at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$), where the joint solution was better by a median of 0.90 in total cost. That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, a cheap early choice propagates and the difference is no longer confined to dense nodes (Section 3.3).

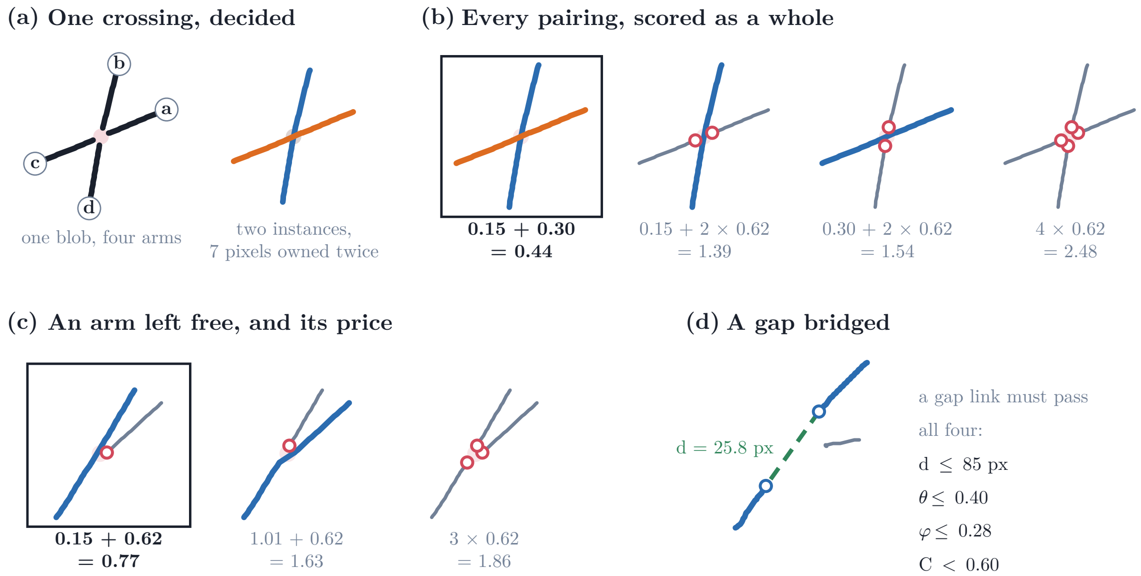

Figure 3. Every decision the linker can make, on the nodes it made them at. Every number shown is measured, and every arm is the skeleton it was measured from, drawn through a short moving average that removes the raster staircase and moves no point by a pixel; the two nodes are rigidly rotated to fit their panels, so every angle shown is the measured one. (a) A four-arm crossing (development scene `cov20/synth_0001`, node 9) before and after the decision: arms named a to d anticlockwise by outward tangent, the junction cluster shaded, and the two instances it becomes, which share seven pixels. (b) Every configuration the admissibility gate allows at that node, each scored as a whole. Under each is the sum of its edge costs and the price $p_i = 0.62$ of every arm it leaves free, a ring marking such an arm; the boxed configuration is the one returned. The two candidate pairs missing here cost 3.15 and 3.33, above $p_i + p_j = 1.24$, so pairing them would cost more than leaving both free. (c) The same at a three-arm node (`cov40/synth_0001`), where an admissible edge costing 1.01 is declined: taking it totals 1.63, against 0.77 for pairing the two arms that continue into one another and paying to leave the third free, and 1.86 for leaving all three free. (d) The third decision, on the same scene: two arm tips 25.8 px apart joined across a break in the mask, with the other mask pieces of the crop in grey. A gap link must clear all four gates at once.

2.4.3 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the fixed JSON configuration.

2.5 Image-informed width reconstruction, optional

The grouping above reads the mask alone and is what this paper evaluates. Where a grayscale image of the same field exists, two further stages can use it to give each instance a width, and both are off by default and governed by limits recorded in the fixed configuration. Neither is specific to SEM: they take whatever image accompanies the mask, and SEM is simply the case demonstrated here.

Perpendicular intensity profiles, sampled along each ordered chain away from junctions, attach an apparent width, a brightness, an uncertainty and a quality flag to an existing instance without altering its arm membership. Ribbon rendering then interpolates accepted gaps, resamples and smooths the centreline, and draws the chain at its fitted width, excluding junction blobs. Silhouette absorption, which would union rendered instances, has threshold zero by default; neither it nor any post-hoc continuation merge is part of the evaluated algorithm or contributed to the held-out result. Because the grouping is fixed before either stage runs, no width measurement can move a pixel from one instance to another, and no identity score in this paper depends on one.

2.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1. The input mask is cut into fragments, each one unambiguous arm between crossings, by a rule that reads neither the reference nor any prediction, so every method compared is cut into the same pieces. Each fragment then takes a reference identity and a predicted identity by one rule, applied to both sides alike: it is assigned to the instance that overlaps most of its pixels, provided that overlap covers at least 30 % of the fragment; failing that, to the most frequent nearest instance label within 6 px; failing both, it stays unassigned and enters the score as its own singleton cluster, so every fragment is in the scored partition and unattached evidence is penalised rather than dropped. The score counts the pairs of fragments the prediction groups as the reference does. The scorer's preprocessing shares two constants with the method itself: both prune skeleton spurs of at most 3 px, and the scorer's 6 px minimum fragment length and assignment fallback distance coincide with the 6 px below which PLECTA omits an isolated component, so method and endpoint rest on a shared preprocessing assumption rather than independent ones. A prespecified one-at-a-time sensitivity sweep of the four constants, re-scoring the stored predictions with the grid frozen before any result was seen, is reported in Appendix A. Because junction pixels are deleted before fragments are labelled, it cannot see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them, which is the difficult decision; Section 3.4 separates the two and measures the first directly.

Several measures accompany it, and the reason for carrying more than one is a single distinction: *pairwise scores relationships between pieces, detection scores the existence of objects.* A filament delivered in n pieces contributes $\binom{n}{2}$ pairs but one object, so pairwise F_1 is quadratic in fragmentation and a broken mask is charged many times over for one missed join, while object detection is blind to an arm swap at a crossing, scoring an inverted reconstruction 1.000 where pairwise scores 0.000; and neither states how many crossings were solved. So the adjusted Rand index and variation of information [34, 35] are quoted beside the pairwise score, detection F_1 beside them, join F_1 where two mask sources fragment differently, and reference-instance recovery to weight every filament equally rather than by its pair count. Crossing Fidelity, the fraction of reference crossings at which the predicted grouping of the incident arms matches the reference partition exactly, says how many crossings were solved, and is an in-house diagnostic rather than an established metric, in the tradition of measures that score correctness at branch points rather than per pixel [36, 37]; it is always printed beside its own all-unpaired baseline, the fidelity of the deterministic do-nothing policy that leaves every arm unpaired. The input masks themselves, as distinct from the reconstructions, are characterised by the tolerant completeness/correctness/quality triple standard for thin structures. Appendix A defines each measure in full, with its equation, what it rewards and what it cannot see; every one is quoted with the others rather than alone.

Statistical analysis

Held-out means and density strata are accompanied by 95% nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810), resampled with stratification by density; the clean-minus-degraded analysis resamples paired scene differences within density. Per-coverage points in the comparison figures are plain means of ten scenes with no interval drawn and none claimed. The real SEM work is exploratory, 3 fields, and no inference is made from it; the inferential unit there is the field, and the 2566 reference crossings entering the crossing measures are a count pooled over six condition-specific evaluations of the three fields, not a sample size. Active-component ablations are one-at-a-time development experiments on all 84 development scenes and do not reuse the held-out set.

3 Results

3.1 Held-out synthetic reconstruction

PLECTA completed all 128 held-out scenes with 0 failures. Mean common-fragment pairwise F_1 was 0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]); Table 1 gives precision, recall, reference-instance recovery, adjusted Rand index and both variation-of-information components, overall and by density. These scenes used unseen seeds from the same procedural generator as development data, so this is same-generator generalisation and not distribution transfer (Section 4.1).

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information, for which lower is better; higher is better for the other metrics.

Group	n	F_1	Precision	Recall	Recovery	ARI	VI split	VI merge	VI total
All	128	0.854	0.868	0.843	0.799	0.851	0.256	0.216	0.473
20%	32	0.911	0.928	0.896	0.918	0.907	0.137	0.093	0.230
30%	32	0.874	0.892	0.858	0.828	0.870	0.211	0.161	0.372
40%	32	0.858	0.864	0.853	0.796	0.855	0.242	0.224	0.466
60%	32	0.775	0.787	0.765	0.656	0.772	0.436	0.388	0.823

Performance decreased with scene density: mean F_1 fell from 0.911 [0.888, 0.933] at 20% coverage to 0.775 [0.760, 0.790] at 60%, monotonically through the intermediate strata. The same decline appears on the paired clean-against-degraded set of Section 3.5, where the clean and the degraded curves separate at low coverage and meet at 50–60 %: degradation costs least exactly where the score is lowest, because there the tangles rather than the gaps set the difficulty. Per-scene runtime had mean 2.54 s, median 1.12 s, 90th percentile 6.42 s, and maximum 34.06 s; these timings are hardware-specific.

3.2 Which density axis matters

Coverage is the fraction of the frame filled by the rendered bundles, so it rises both when filaments are added and when the same filaments are drawn thicker, and only the first changes the grouping problem. A factorial separates them: twelve geometries, three centreline length densities crossed with four seeds, were each rendered at bundle widths of 6, 11 and 16 px. Within one geometry the input axis mask and the overlap-aware centreline reference are byte-identical across the three widths, verified by SHA-256, while the union of the thick supports changes: areal coverage varies by up to a factor of 2.49. Any movement in the score across those three renderings would mean that the method or the metric reads the silhouette.

Across all 12 geometries the within-geometry F_1 range was 0.0000, that is, exactly zero in 12 of 12 (Table 2). Areal coverage therefore cannot causally affect the grouping score; PLECTA reads the axis mask, and the metric scores centreline fragments, so neither sees the width the bundles were drawn at. Centreline length density does move it, mean F_1 falling from 0.920 to 0.836 across the three densities of Table 2. The coverage bands of the three length densities also overlap, so coverage does not identify a difficulty level on its own. Density strata reported by target areal coverage should therefore be read as a proxy for the length density accompanying them under this generator.

Table 2. Bundle width against centreline length density, 36 development scenes. Within each row the three bundle widths share one byte-identical axis mask and one reference, so the spread in areal coverage carries no score difference. Length density is centreline pixels per image pixel.

Centreline length density	n	Areal coverage	Bundle width	F_1
0.0193	12	0.118–0.309	6, 11, 16 px	0.920
0.0372	12	0.218–0.505	6, 11, 16 px	0.878
0.0548	12	0.305–0.644	6, 11, 16 px	0.836

3.3 Four linkers on identical masks

One baseline and two published methods were run on identical input masks with one scorer, over 50 scenes generated after every configuration was fixed. The published methods are DNAi [26] and GraFT [27], each executed from its own released source. All were scored through the same conversion and metric code as PLECTA, and that path was gated on parity: before any comparator number was believed it had to reproduce PLECTA's stored F_1, adjusted Rand index and both variation-of-information components, because a faulty conversion yields plausible rather than obviously wrong scores. Each comparator received a parameter translation selected on the same 84 development scenes PLECTA was configured on, and every adaptation was checked and found to favour the comparator; Appendix B records each one, what it was worth, and how each method fails.

Greedy continuation. The baseline skeletonises the same mask and greedily accepts the lowest-turning-angle pairing between branch ends inside a distance and a turning-angle gate, both fixed by a development sweep. It is ours, not a reimplementation of any particular paper but of the pairing principle SIFNE, GraFT's path cover and the continuity baseline of Wegmayr et al. [23, 25, 27] share, and it is reported as the transparent floor the claim of Section 1 is made against, never as an external result. PLECTA leads it by +0.145 F_1 [+0.129, +0.162] paired over the 50 scenes, and by +0.128 [+0.113, +0.144] on clean axes, so the margin is not an artefact of mask degradation.

Table 3. All four methods on the degraded masks, identical input masks and one scorer. $n = 50$ scenes and 3638 reference crossings, except GraFT, over the 47 it completed. Variation of information is in bits and lower is better; higher is better for the rest. Crossing Fidelity is defined in Appendix A, and leaving every arm unpaired scores 0.088 on it here. Paired differences with their intervals are in the text.

Measure	PLECTA	Greedy	DNAi	GraFT
Common-fragment F_1	0.822	0.676	0.343	0.490
Precision	0.831	0.785	0.462	0.666
Recall	0.814	0.599	0.284	0.400
Reference-instance recovery	0.765	0.542	0.252	0.304
Adjusted Rand index	0.817	0.670	0.331	0.479
Variation of information ↓	0.573	1.036	2.107	1.558
Crossing Fidelity	0.809	0.729	0.581	0.683
Median s per scene, 20/60%	0.14–4.1	—	0.04–0.6	2.44–29.7

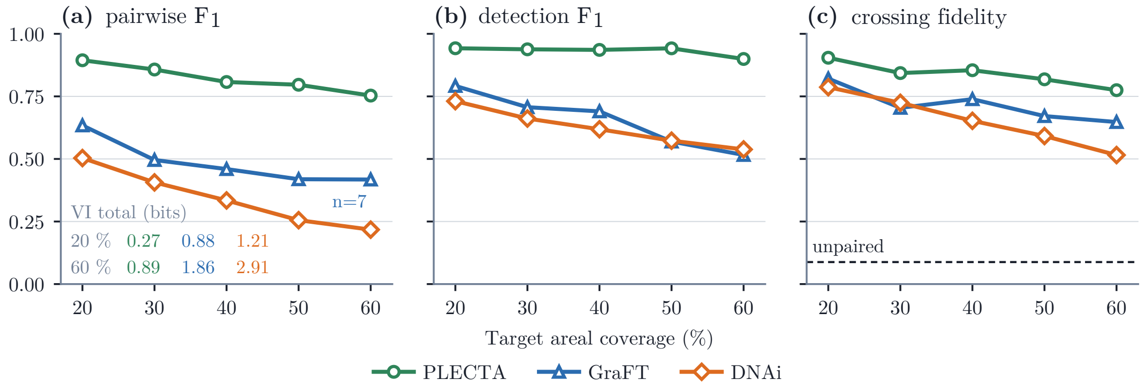

Figure 4. The three methods on the degraded input masks, by coverage stratum, under three of the measures of Section 2.6: **(a)** pairwise F_1, which counts pairs of fragments; **(b)** detection F_1 at IoU 0.1 with the 2 px placement tolerance, which counts objects; **(c)** Crossing Fidelity, which counts crossings, against the fidelity of leaving every arm unpaired (dashed). The panels share one 0–1 scale, with tick labels written once. The numbers inside (a) are each method's variation-of-information total in bits at the two ends; that measure has no panel because PLECTA's whole trajectory lies under 0.9 bits against DNAi's 2.9 and a shared axis crushes it. Points are plain means of $n = 10$ scenes, no interval drawn and none claimed; the paired differences are in Table 3. GraFT exceeded its wall-clock budget on three of the densest scenes, the hardest of the set, so its rightmost point is a mean of seven and is optimistic in all three panels.

Exact against greedy matching, matcher only. That baseline replaces the whole linker; a second arm isolates the claim, keeping the graph, cost, admissibility gate and round schedule and replacing only the per-junction maximum-weight matching by greedy cheapest-first acceptance. Exactness was worth +0.080 F_1 [+0.070, +0.089] over 30 development scenes, on 29 of 30 of them, and +0.090 [+0.080, +0.100] on clean axes; the failure is division, not merging, with variation-of-information split rising 0.252 to 0.500 at unchanged merge, 30 % more instances emitted and crossings resolved exactly falling 0.861 to 0.668. The relative loss grows monotonically with degree, -21 % at degree three to -35 % at degree six, but in absolute terms peaks at degree four (-0.252), where most crossings lie; and the greedy arm runs at parameters selected with the exact matcher in the loop, a bias in that matcher's favour.

DNAi. Its segmentation network was bypassed and only its instance-forming stage used, so grouping is not confounded with segmentation quality, and it was run at a development-selected configuration rather than at its released defaults. That adaptation is worth a great deal to it: at those defaults it scores F_1=0.195 on these masks against 0.343 tuned. Tuned, the paired difference is +0.478 [+0.456, +0.501] over 50 scenes, and +0.414 [+0.392, +0.436] on clean axes, the gap widening monotonically with density (Figures 4 and 5).

DNAi's margin is a limitation of domain and density, not a general ranking: it is built for DNA-fibre micrographs, which are sparse, and where the network is sparse and the axis clean it recovers 0.660 at 20% coverage. Restricting the score to the evidence a degree-limited, gap-blind method can reach separates capability from judgement, and on clean axes the difference falls from +0.414 to +0.284 [+0.262, +0.307], still favouring PLECTA in 49 of 50 scenes and at every coverage level. Roughly 31 % of the margin is therefore coverage of cases the comparator cannot attempt, and 69 % the outcome of decisions both methods make.

GraFT. Its published entry point begins with a Frangi and Otsu front end for grayscale ridge images, which has nothing to do on a two-valued axis, so the mask was injected at its skeleton-to-graph stage, everything downstream being its own code unchanged. That substitution favours it: run end to end on the grayscale images it scores 0.100, because its segmentation stage recovers only a third to a half of the reference skeleton. Its minimum continuation angle was also re-tuned on development scenes. GraFT scored 0.490, a paired difference of +0.339

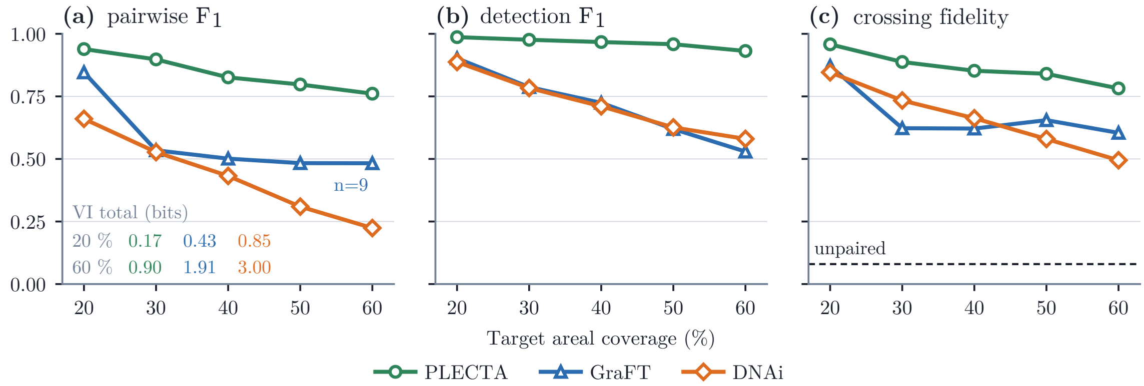

Figure 5. The same three methods, the same scenes, the same scorer and the same panels as Figure 4, from the generator's undegraded axis masks instead of the degraded ones. Every method scores higher and the order does not change at any coverage under any of the three measures. One of GraFT's densest scenes was censored here rather than three.

[+0.303, +0.378] favouring PLECTA in every one of the 47 scenes both methods completed, and +0.275 [+0.224, +0.329] on clean axes. 3 of the densest degraded scenes did not finish inside a 45-minute budget and are excluded from its means; they are the hardest scenes of the set, so that exclusion flatters GraFT. Neither released method establishes a general ranking, and Section 4.1 states what these comparisons leave untested.

A second generator. Every scene above comes from one generator, so the comparison was repeated on GraFT's own, run unmodified at its published settings over 40 scenes in 8 density strata spanning 0.0016 to 0.0557 centreline pixels per image pixel; GraFT's published comparison reaches 0.020, so its whole range is the lower half of this one. PLECTA leads at every density, from 0.970 against 0.787 and 0.659 at the sparsest to 0.800 against 0.219 and 0.149 at the densest; the margin widens with density and nothing crosses over, including inside GraFT's published range. The field-standard measures order the three identically over the whole ladder (adjusted Rand index 0.912, 0.453 and 0.319; variation of information 0.262, 1.629 and 2.033 bits), and PLECTA's errors lean mildly to fusion where both comparators divide far more than they fuse. GraFT was also run end to end with its own front end, which on this data is inside its design envelope rather than degenerate as it is on ours; that arm scored 0.609 and 0.237 at the two ends, below the injected arm almost everywhere, so injecting the mask continues to favour it. The held-out result of Section 3.1 is a same-generator measurement and remains one; this comparison is not.

Crossings resolved. Panel (c) of Figures 4 and 5 states the contribution directly, and pooled over the 3638 reference crossings of the degraded set PLECTA partitions 0.809 of them exactly against 0.729 for the greedy baseline, 0.683 for GraFT and 0.581 for DNAi; paired, PLECTA's margins are +0.065 [+0.047, +0.083], +0.123 [+0.085, +0.165] and +0.185 [+0.158, +0.211]. Clean axes order them the same way, PLECTA at 0.827. Leaving every arm unpaired resolves 0.088 of these crossings, which is the all-unpaired baseline the rate is read against; it is a deterministic do-nothing policy rather than a chance level. It is much less discriminative than the pairwise score, spanning 0.259 between best and worst method at 60 % coverage where pairwise F_1 spans 0.536, so it is used to say what was solved and not to rank. Everyone is poor at the busiest nodes: above degree four PLECTA resolves 0.414 and no comparator does better, while DNAi collapses to 0.016, which is what its degree-four cap predicts.

Per-scene cost was measured warm, single-threaded and pinned to one core, timing each method's own computation only (Table 3; 3 GraFT scenes exceeded 600 s and are censored). These are independently optimised codebases, so absolute times measure implementation as much as algorithm; the transferable quantity is scaling, and the log–log exponents against foreground pixels are 2.11, 2.65 and at least 2.01. PLECTA has the largest uncensored exponent; GraFT's is a lower bound, because the scenes its fit lost to censoring are the slowest ones, and only GraFT is heavy-tailed.

3.4 Crossing representation, and where it over-marks

Resolving a crossing and representing one are different things, and only the first is what the scores above measure. Which arm continues into which is the difficult decision, and every score in this paper rewards getting it right. Whether the emitted raster *represents* a crossing pixel as belonging to both filaments is a property of the output format, and the metric cannot see it: flattening the emitted layers to a single label image leaves PLECTA's F_1 unchanged to four decimal places (0.0000).

The representation is instead verifiable directly: on the comparison set, reference instances share 11 % of their pixels with another instance at 20% coverage, rising to 39 % at 60%; PLECTA emits 5–20 %, and GraFT, whose output is close to a hard partition, about 2 %. Which pixels those are is less flattering. PLECTA recovers 0.463 of the pixels the reference gives to more than one instance, less than half, though the largest such recall here, against 0.144 for GraFT and 0.128 for DNAi, while the greedy baseline, a hard partition, scores zero by construction. It does so at a precision of 0.134, marking 3.6–4.0 times more shared pixels than the reference contains, because a chain is painted with every pixel of every junction cluster it touches. That is a property of the rendering rather than of the grouping, and one the F_1 neither rewards nor penalises.

Figure 6 shows what the score corresponds to scene by scene, taking at three coverage levels the scene whose F_1 is closest to the median, so the rows are typical rather than selected.

3.5 Response to the input mask

On the separate 50-scene controlled set, mean F_1 was 0.822 from degraded axes and 0.844 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113, 0.0333]). Modelled gaps and raster irregularities therefore caused a small resolved loss under this generator.

The thickness of the axis mask costs more, and it matters because it is the upstream segmenter, not the method, that fixes it: over 20 development scenes a nominal 3 px axis mask scored -0.047 F_1 [-0.073, -0.023] against the 1 px mask of the same scenes. Each width variant is scored against common fragments rebuilt from its own input mask, so the question asked is how well the method groups the skeleton it is handed, and the decline with width is a thicker mask skeletonising into a messier graph rather than a change in the grouping problem. This is the one density axis that does move the score. The rendered silhouette does not, causally and scene by scene (Section 3.2), so a mask should be reported by the geometry of its axis and not by how much of the frame its bundles fill.

3.6 Development ablations and downstream rendering

One-at-a-time ablations were evaluated only on the 84 development scenes (Table 4). The three largest losses relative to the evaluated configuration were, in order, the junction chord-turn term (-0.1858), gap matching (-0.1394) and connector-based junction-cluster consolidation (-0.0634); every other component, chain-extended frames and the round schedule included, mattered less. These are development values, not held-out effect estimates.

The core output is a grouping, not a set of connected curves: accepted gap links establish identity but are not painted, so on the development set 0.653 of core instances arrive as more than one disjoint pixel run and 0.663 contain a branch point. Any downstream step that assumes one connected curve per instance must account for this. The optional ribbon rendering of

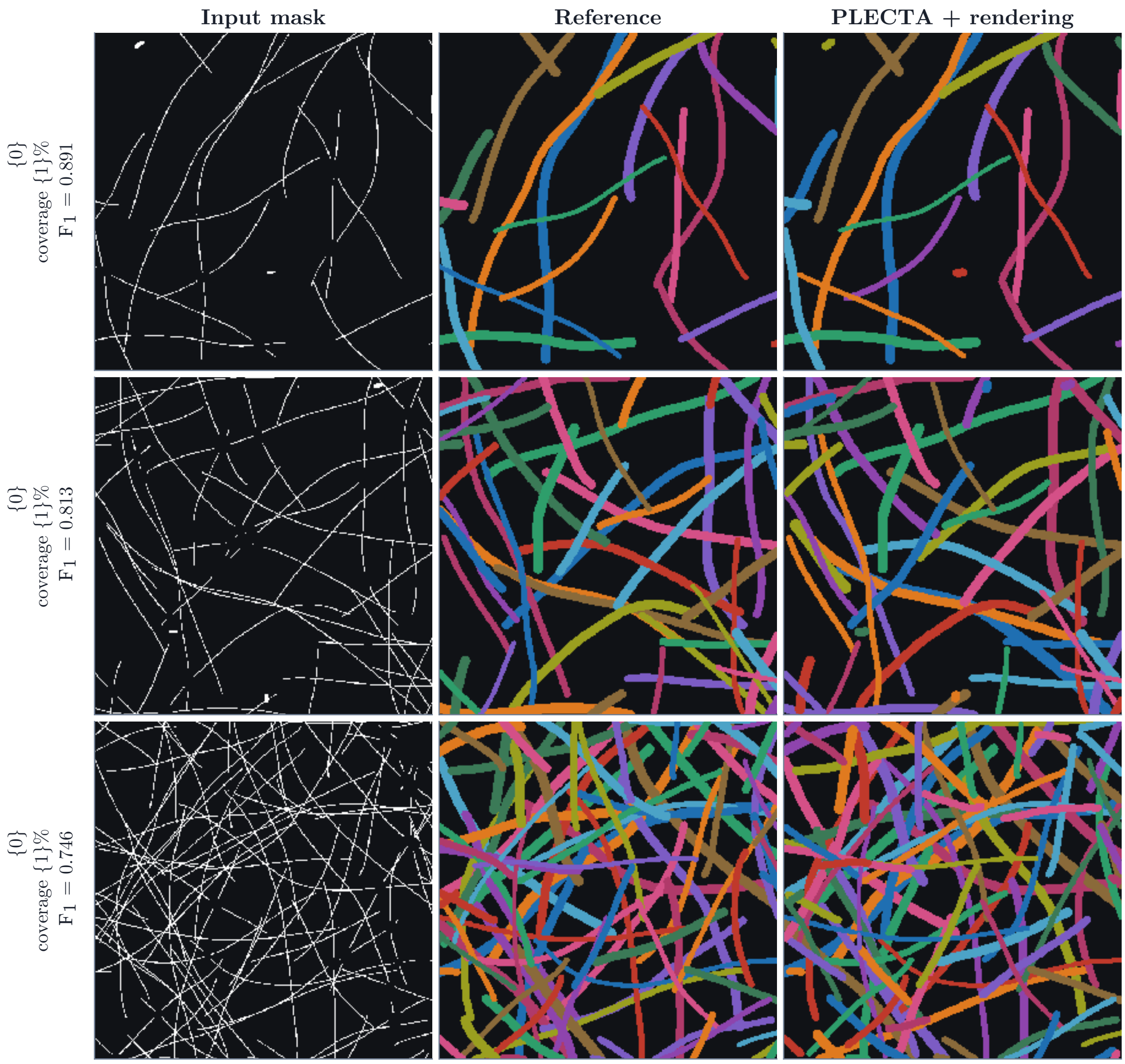

Figure 6. Reconstruction at three coverage levels, each the scene closest to the median F_1 of its stratum. Columns: the degraded input mask, the overlap-aware reference, and PLECTA's instances drawn at their fitted widths. What is evaluated in this paper is the grouping of centreline pixels the method produces from the mask alone, and the F_1 beside each row is that grouping's score; the width rendering shown here is a downstream step that redraws those instances and cannot move a pixel from one to another. Instance colours are arbitrary and not matched between columns.

Section 2.5, at its defaults, scored development F_1=0.860, a change of -0.0068, and reduced those fractions to 0.000 and 0.005, so rendering buys connectivity at a small cost in the identity score without changing chain membership. Enabling the separate 0.6 silhouette-absorption rule reduced F_1 further to 0.857, and because absorption can merge identities it remained disabled. None of these topology measurements validates apparent width.

3.7 Proof of concept on real SEM

The claim of this section is one sentence: PLECTA works on real imaging when a second technique supplies the mask, and Pipeline A is such a technique. Nothing here is a measurement of Pipeline A, which carries its own evaluation [33].

Pipeline A runs procedural mask → CycleGAN → paired synthetic corpus → nnU-Net → binary filament-axis mask: a generator draws a mask, a CycleGAN translates it into an SEM-like

Table 4. One-at-a-time component ablations on the 84-scene development set. Negative ΔF_1 values denote losses relative to the evaluated configuration.

Development variant	F_1	ΔF_1	ARI
Fixed configuration	0.867	+0.000	0.864
No gap bridging	0.728	-0.139	0.722
Single round	0.847	-0.020	0.844
No chain-extended frames	0.846	-0.021	0.842
No round schedule	0.863	-0.004	0.860
No curvature term	0.863	-0.005	0.859
No short-connector node merging	0.845	-0.022	0.841
No junction-cluster consolidation	0.804	-0.063	0.799
No junction chord-turn term	0.681	-0.186	0.675

image so that image and mask are paired without an annotator, an nnU-Net trained on those pairs segments a real micrograph, and its output is a binary axis mask [32, 33]. Pipeline B is PLECTA, and it is the only stage evaluated here. 3 manually annotated CNT-sheet fields separate the two: PLECTA is run once on the skeleton of the manual annotation and once on the nnU-Net axis, with the annotation as the reference in both cases, so the two conditions differ only in the mask. The first is an oracle-like input condition rather than an end-to-end result, asking what the grouping can do when the axis evidence is curated; only the second says what an automatic mask allows. Figures 8 and 7 show both on one field, at width and as the centrelines the scores are computed on.

(a) Manual-derived axis **(b)** nnU-Net axis

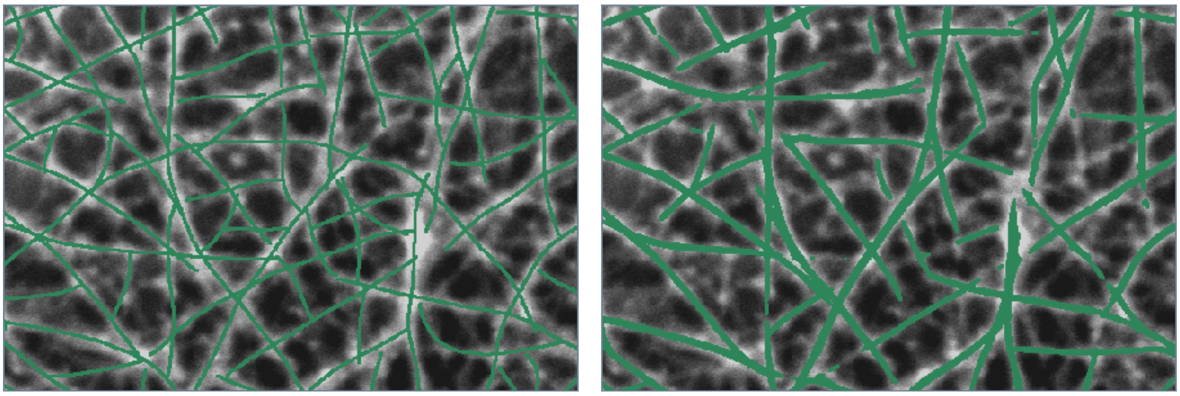

Figure 7. The two input axes of field B58_110, drawn over the micrograph they were read from: **(a)** the skeleton of the manual annotation and **(b)** the output of Pipeline A. These are the binary masks themselves, before any reconstruction, so there is nothing to tell apart within a panel and one colour is sufficient; the comparison the figure makes is between them. Everything separating the two conditions of Table 5 is visible here: the same filaments, read a second time by a network rather than by a person, arrive thicker, broken into more pieces, and displaced by a pixel or two almost everywhere. The axis is dilated by one pixel for legibility at this size, which changes no geometry. B58_300 behaves the same way, as its mask-quality triple in the text records.

Which fields are held out, and what the exception costs. B58_110 and B58_300 are absent from the training set of the inference run used here, verified by image content rather than by filename at tile correlations of 0.10 and 0.15, so both are genuine held-out observations of mask transfer. B58_100's image is present in every available training run at 0.9999 or above and cannot be made otherwise; its U-Net row is an upper bound and is marked as such in Table 5. What that costs is measured rather than left vague. Running 3 nnU-Net models over all three fields scores the same field under models that held it out and one that did not: on B58_110 the two clean models give 0.451 and 0.453 against 0.512, on B58_300 0.464 and 0.468 against 0.537.

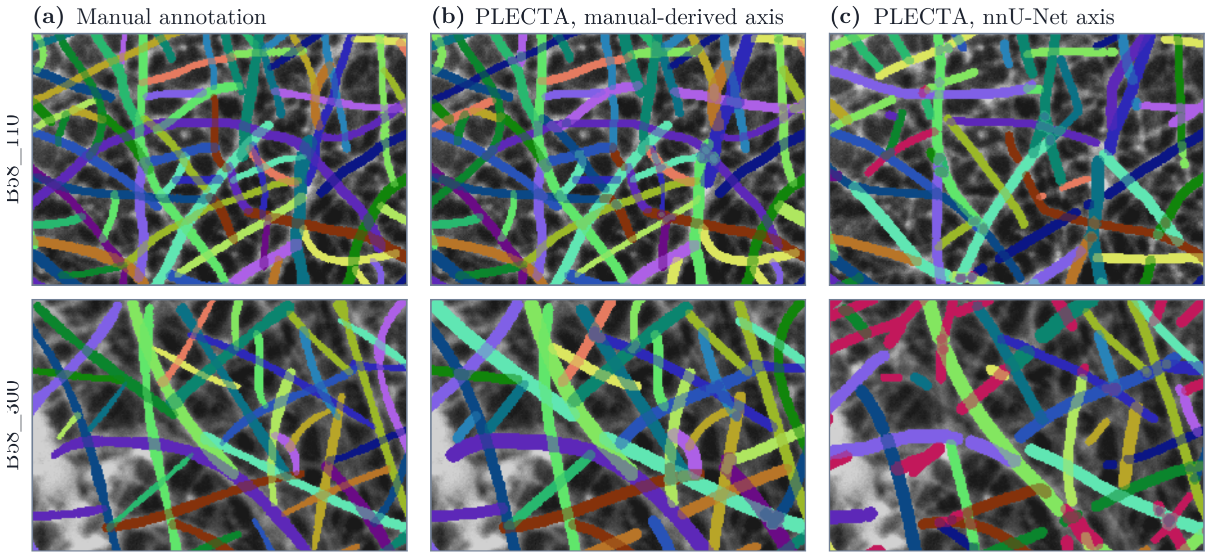

Figure 8. Reconstructed bundles on the two SEM fields held out of the nnU-Net run that produced their axes, one field per row. **(a)** the manual annotation; **(b)** PLECTA from the manual-derived axis; **(c)** PLECTA from the nnU-Net axis of Figure 7b, the same method and the same parameters, differing only in the mask. All three are drawn at width so that they are the same kind of object: the annotation as the annotator painted it, and the two reconstructions through the optional image layer of Section 2.5, which measures a width per instance across the centreline and renders a ribbon. That layer runs after the grouping is fixed and cannot move a pixel from one instance to another, so no score in this paper depends on it. Instance colours correspond across the panels of a row: a predicted instance takes the colour of the annotated filament it was matched to, one matched to none is crimson, and adjacent filaments are given different colours, which carry identity within the row and nothing else. Each crop is the window whose local pairwise F_1 is jointly closest to that field's whole-field value on both axes, so it is typical of the field rather than selected from it.

Training on a field is worth 0.060 to 0.071 pairwise F_1, the trained-on model predicting about a third more foreground, which is the mechanism. On B58_100 every model trained on the field, the three agree, and there is no gap to see: that is the control.

Table 5. Reconstruction from the manual-derived and the nnU-Net axis in three CNT SEM fields, under the measures of Section 2.6. *Held out* is whether the nnU-Net run that produced the axis excluded that field from training; B58_100 could not be held out by any available run, so its U-Net row is an upper bound. *CF* is Crossing Fidelity and *Unpaired* the fidelity of the all-unpaired baseline, the do-nothing policy that leaves every arm unpaired on that field. *Cross.* is the number of reference crossings scored. Detection F_1 is at IoU 0.1 with the 2 px placement tolerance. Exploratory, 3 fields, no inference.

		Held	Grouping			Objects and crossings			
Field	Input axis	out	F_1	Join F_1	Decisions	Det. F_1	CF	Unpaired	Cross.
B58_100	Manual-derived	–	0.887	0.929	13 234	0.941	0.922	0.013	371
B58_100	U-Net	no	0.457	0.559	17 282	0.633	0.617	0.102	384
B58_110	Manual-derived	–	0.837	0.911	24 870	0.930	0.920	0.023	599
B58_110	U-Net	yes	0.452	0.593	16 134	0.602	0.614	0.048	290
B58_300	Manual-derived	–	0.947	0.959	28 224	0.959	0.956	0.011	611
B58_300	U-Net	yes	0.466	0.615	16 729	0.577	0.617	0.048	311

What the three fields say. From the manual-derived axis PLECTA reached F_1=0.887, 0.837 and 0.947, and resolved 0.922, 0.920 and 0.956 of the reference crossings exactly against an all-unpaired baseline of 0.011 to 0.023. From the nnU-Net axis the same predictions read 0.452 and 0.466 on the two held-out fields, with Crossing Fidelity 0.614 and 0.617 and join F_1 0.593 and 0.615. The two clean fields agree with each other to within 0.03 on every measure, which is the substantive gain over the single clean field the study previously had; 3 fields still cannot estimate a population, and the inferential unit is the field: the 2566 reference crossings the crossing measures rest on are pooled over six condition-specific evaluations of these three fields, a pooled count and not a sample size.

Most of the distance between the two conditions is the mask, and the method fails by conservatism rather than by fusing. The axis each field hands PLECTA is characterised, before any reconstruction is opened, by the tolerant completeness/correctness/quality triple: at 2 px, quality is 0.504, 0.557 and 0.507, over completeness 0.671, 0.651 and 0.619. The tolerance is not a formality: quality falls to 0.299, 0.375 and 0.325 at 1 px, because the annotation and the nnU-Net output are two independent readings of the same filaments and disagree in placement almost everywhere. That displacement is also why detection F_1 carries a placement tolerance, and the control that makes it a correction rather than an inflation is here: on the manual-derived axes, where the input mask is a skeleton of the annotation itself, the tolerance moves detection F_1 by 0.000, against 0.179 rising to 0.602 on B58_110's nnU-Net axis. Because the pairwise score is quadratic in fragments, a filament the nnU-Net mask breaks into several pieces carries every one of its pairs; join F_1, which counts one decision per pair of fragment ends the mask brings within reach, reports 24870 decisions on B58_110's manual axis against 16134 on its nnU-Net one. It narrows the gap without closing it, and its decomposition says what the rest is: a join precision of 0.681 against a recall of 0.525 is a method declining joins rather than fusing filaments, which is what panel (c) of Figure 8 shows.

4 Discussion

PLECTA addresses a narrow problem: assigning fragments of a binary filament-axis mask to overlapping projected instances. Explicit geometry recovers substantial identity there with no instance-labelled training data and no image intensity, and what recovers it is the combination of local and accumulated evidence, neither of which suffices alone: a junction is resolved by mutual compatibility rather than by the cheapest available continuation, and the evidence for that judgement is re-read along the chains as they assemble. That evidence is directional throughout, which fixes the condition the image must meet: a tangent measured on one side of a crossing has to predict the arm leaving the other, which bending stiffness produces most directly and tension, confinement or bundling can produce as well. The ablations are consistent with that reading, the two largest losses being the junction chord-turn term and gap matching, both purely geometric. Where a filament is flexible on the scale of a junction the premise fails rather than degrades: there is then nothing local to match, and the correct response is to acquire evidence the projection has lost rather than to weaken the gates. The ablations locate that effect without proving each term independently necessary: the two large losses correspond to the two kinds of discontinuity, the junction and the missing mask segment, while accumulated context is worth distinctly less than either.

Difficulty follows the density of centrelines rather than the amount of ink, so what costs the score is the number of locally plausible alternatives at a crossing. The rest of the sensitivity belongs to the upstream segmenter, which fixes the thickness of the mask and therefore how clean a skeleton PLECTA is given. PLECTA can bridge a short omission, but it cannot infer a filament absent from the mask, nor reliably reject a false axis introduced upstream. Supplying identical masks isolates grouping in an algorithmic comparison; it does not establish end-to-end SEM performance, which is why the three-field case is a demonstration of the joined pipelines rather than a measurement of them. Ribbon rendering, being downstream of grouping, buys connectivity rather than identity, because the core paints arms and junction clusters but not the gap links that carry identity across a break; width and brightness can be reported for characterised instances, but their physical accuracy requires separate references.

4.1 Evidence boundary

The principal limitation is distributional, and it is narrower than a one-generator study. The held-out scenes use independent seeds but the same procedural generator family used for development, so their confidence intervals quantify sampling variation within that family and not transfer to another simulator, specimen, microscope, or segmentation pipeline. The *ranking*, however, does not rest on that family: GraFT's own generator was run unmodified at its published settings over 40 scenes in 8 density strata, and its bundled frame separately, and the ordering held on both (Section 3.3). What is untested is transfer of the held-out *level* to a third generator, not of the comparison. The ablation and rendering studies are development-only and remain secondary to the fixed held-out result.

The real SEM study is deliberately a case study with 3 fields, not a population validation. Manual-derived masks are an oracle-like input condition: they test grouping when the axis evidence is curated, not automatic SEM-to-instance performance. B58_100 is training-contaminated for the U-Net condition and cannot be made otherwise, so its row there is an upper bound; B58_110 and B58_300 are held out, and two fields that agree closely are a stronger statement than the one the study previously had, but they still cannot estimate generalisation. The two mask sources also generate different fragment universes, so their F_1 values describe performance under each input condition rather than comparing identical objects. More independent fields, sealed before configuration, and multi-reader annotations are required.

One caveat from the comparisons must be separated from the claim it seems to threaten. The comparators lose on how a crossing is *resolved*, which the metric scores and which is the difficult decision, not on how it is *represented*, which the metric cannot see: GraFT's near-hard partition is a real limitation that costs it nothing here. Both properties are worth having, and Section 3.4 measures the representation directly, including where PLECTA over-marks it.

Two further limits concern the output itself. PLECTA recovers projected two-dimensional identity: shared pixels encode overlap but not which filament passes above another, so no three-dimensional topology follows. And the grouping metric evaluates ownership of centreline fragments, which is not a validation of rendered width; quantitative diameter claims require matched physical measurements across the relevant scale, contrast, and specimen ranges. Pairwise fragment scores also emphasise instances with many fragments, which is why recovery and the adjusted Rand index are reported alongside.

The comparisons themselves are bounded. The greedy rule is our reimplementation of a principle rather than of any particular paper, and DNAi was run outside the domain it was built for, part of its cost function reading a colour channel our masks do not have. What the results support is narrow and mechanical: pairing endpoints greedily, or committing to a pairing with no option to decline, loses identity in dense crossings. They do not support a claim that PLECTA is generally superior; that would need several released implementations run in their own regimes by their own authors. Broader validation should extend to independent real masks and other filament domains, with input fragments and scoring rules held fixed.

5 Conclusions

PLECTA reconstructs overlapping filament instances from a binary axis mask by alternating exact junction matching with gated gap matching as local arm geometry expands into chain-level evidence. Its evidence is directional throughout, so it requires tangent and curvature to persist across a crossing at the scale of its fitting support, a condition bending stiffness contributes to and rod-like filaments commonly meet. That is the regime of nanotube bundles and of the cytoskeletal and synthetic-fibre networks comparable methods target, and not that of freely flexible chains, where the projection retains nothing local to match. The fixed method achieved common-fragment $F_1 = 0.854$ on 128 independently seeded held-out scenes from the same procedural generator, with performance declining as network density increased, and partitioned 0.809 of the comparison set's reference crossings exactly as the reference does. On a generator that is not ours, being GraFT's own, run unmodified at its published settings over 40 scenes in 8 density strata, it reached $F_1 = 0.916$ against 0.475 and 0.349 for the two released comparators, so the ordering does not rest on the generator that produced the held-out set. On identical masks it gained +0.145 F_1 over a greedy minimum-turning-angle continuation rule, the rule shared by several published linkers, and +0.478 and +0.339 over two released implementations run from their own source; these support solving each junction exactly rather than pairing endpoints in priority order, but the released comparisons are bounded by domain and density and do not establish a general ranking. Development ablations identify junction chord-turn evidence, gap bridging, and junction-topology consolidation as the principal contributors, and optional image-informed width reconstruction improves geometric presentation while remaining separate from the grouping claim.

The proof of concept on carbon-nanotube sheets joins the two pipelines: an upstream synthetic-data-driven segmenter prepares the mask, and PLECTA resolves it into instances, demonstrated on 3 annotated SEM fields under manual-derived and nnU-Net masks, two of them held out from that segmenter, without establishing population generalisation or width accuracy. The method therefore offers an inspectable route to projected filament identity that draws on no random number generator and, in its grouping core, no training data; the demonstrated SEM route is not training-free, since the mask it consumes comes from a trained CycleGAN and nnU-Net upstream. Broader claims require independent real fields, validated physical widths, further released implementations run in a matched comparison, and an external-generator benchmark with sealed seeds.

CRediT author contribution statement

[To do: Fill in each author's CRediT roles before submission, checked against actual contributions; keep the author names and order as listed above.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at `https://github.com/Jorallan/PLECTA`. Generated scenes, detailed per-scene records, and development diagnostics are retained locally. **[To do: Create a tagged archival release and add its DOI before submission; do not infer or preassign a DOI.]**

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A Measures

Every measure this paper reports is defined here, with what it rewards and its blind spot; none reads the rendered width of an instance.

The common-fragment set, shared by the first five measures below. The mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded, so one fragment is one unambiguous arm between crossings. The partition reads neither the reference nor any prediction, so every method compared is cut into the same fragments. Each then takes a reference and a predicted identity by one rule, applied to both sides alike: the fragment is assigned to the instance whose mask overlaps the largest number of its pixels, the vote counted against each instance's full mask rather than a flattened label image, so an instance that overlaps another keeps its pixels, provided that the winning overlap covers at least 30 % of the fragment; failing that, it falls back to the most frequent nearest instance label within 6 px of the fragment, read from a deterministically flattened field; failing both, it stays unassigned and is scored as its own singleton cluster, on the reference side and the prediction side equally, so every fragment enters the scored partition. Ties at any step resolve to the lowest instance id, and the flattening order is fixed, so the assignment is deterministic. Because junction pixels are deleted first, no fragment contains one: none of these measures can see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them.

Sensitivity to the scorer's constants. Because the four constants above are shared with the method (Section 2.6), the stored predictions of all four methods were re-scored under one-at-a-time perturbations of each constant, with the grid frozen in writing before any result was computed: spur pruning 2–4 px, minimum fragment length 4–8 px, overlap threshold 0.20–0.50, fallback distance 3–9 px. Nothing was re-run and no configuration was changed in response. Table 6 reports PLECTA's mean F_1 and its plain paired margins over each comparator at every perturbation, over the 50 degraded comparison scenes. Across the 10 combinations PLECTA's mean F_1 spans 0.713–0.859, and the smallest paired margin over any comparator under any perturbation is 0.115. The level moves only with the minimum fragment length, and in the same direction for every method, because admitting shorter fragments adds hard, low-evidence units for everyone; the other three constants move the mean by less than 0.002, and no perturbation changes the ordering of the four methods.

Table 6. Scorer-constant sensitivity. Stored predictions re-scored under one-at-a-time perturbations of the four common-fragment constants; Δ columns are plain paired mean margins, PLECTA minus comparator, over the scenes both methods completed. Sensitivity reporting only, no inference, and the published constants were not changed in response.

Scorer constants	PLECTA F_1	Δ greedy	Δ DNAi	Δ GraFT
published constants	0.822	0.145	0.478	0.339
spur prune 2 px	0.822	0.145	0.478	0.339
spur prune 4 px	0.822	0.145	0.478	0.339
min fragment 4 px	0.713	0.115	0.388	0.267
min fragment 8 px	0.859	0.149	0.507	0.358
overlap threshold 0.20	0.822	0.145	0.478	0.339
overlap threshold 0.40	0.822	0.145	0.478	0.339
overlap threshold 0.50	0.822	0.145	0.478	0.339
fallback distance 3 px	0.824	0.145	0.478	0.340
fallback distance 9 px	0.821	0.145	0.478	0.340

Common-fragment pairwise $\mathbf{F_1}$, the primary endpoint. A pair of fragments is a true positive when it shares both identities, a false positive when the prediction joins fragments of different reference filaments, a false negative when it separates fragments of one:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}. \tag{2}$$

A clustering measure of the Rand family [38–40], permutation-invariant in the instance labels. It rewards holding a filament together across every crossing and gap it passes. Blind spots: it is quadratic in fragmentation, so a filament arriving in n pieces carries $\binom{n}{2}$ pairs and a broken mask is charged many times over for one missed join; and it weights long, much-crossed filaments far above short ones.

Adjusted Rand index [34] counts the same pairs corrected for chance, which matters because with a hundred filaments per scene almost every pair genuinely belongs to different filaments and the uncorrected index sits near 1 for anything. Blind spot: redundancy, since over the comparison set it never departs from pairwise F_1 by more than 0.019, so it confirms rather than adds.

Variation of information [35], in bits, lower better, is the conditional-entropy distance between the two partitions: $\mathrm{VI}_{\mathrm{split}} = H(\text{prediction} \mid \text{reference})$ is paid for cutting one filament into several and $\mathrm{VI}_{\mathrm{merge}} = H(\text{reference} \mid \text{prediction})$ for fusing several into one. Reporting both is what separates a method that divides from one that fuses. Blind spot: each component alone is degenerate, so neither is ever quoted singly. A method that merges everything never splits anything and scores a perfect zero on the split component while being useless.

Detection $\mathbf{F_1}$, the measure DNAi reports, matches predicted to reference *objects* greedily by highest intersection over union, stopping below 0.1, DNAi's own shipped threshold, and takes precision over predicted and recall over reference objects. At 0.5 the same predictions read a fragmented mask more than twice as harshly as the pairwise score, which defeats the purpose of carrying an object measure beside it. It uses a symmetric 2 px placement tolerance, under which a reference pixel counts as found when the prediction passes within it and vice versa, the intersection being the smaller of the two counts. The tolerance is needed because two independently read one-pixel centrelines two pixels apart intersect in *nothing*, so a strict overlap reports placement rather than shape, a problem the thin-structure benchmark FISBe states as a design rationale [41]. Counting objects makes it the most forgiving reading of a fragmented mask and the least discriminative measure here. Blind spot: an arm swap, which it scores 1.000 where the pairwise score gives 0.000.

Crossing Fidelity states the contribution directly. At every junction node the reference induces a partition of the incident arms, two together exactly when the reference gives them the same filament, and so does the prediction; Crossing Fidelity is the fraction of crossings at which the two partitions are identical. Nodes are the junction clusters of the same pruned skeleton, and a node's arms are the fragments 8-adjacent to it. Only nodes whose arms carry at least two distinct reference filaments are scored: a node where every arm belongs to one filament is a skeletonisation artefact, and scoring it would reward doing nothing. It is an in-house diagnostic and not an established metric, in the tradition Section 2.6 places it in. Blind spots, and both govern its use: it is far less discriminative than the pairwise score, so it is quoted to say what was solved and never as a ranking axis; and crossings carry no information about *gaps*, so it is silent on the cost the pairwise score pays for bridging them. Sharing its skeleton and assignment with the pairwise score makes it a sharper view of the same evidence, not independent confirmation, and it is always printed beside its all-unpaired baseline, the fidelity of the deterministic do-nothing policy that leaves every arm unpaired.

Join $\mathbf{F_1}$ counts decisions rather than pairs, for the real fields where the two mask sources fragment differently. A *decision* is a pair of distinct fragments whose endpoints come within 85 px, the gap this method will bridge; the reference says join or not, so does the prediction, and precision, recall and their harmonic mean are taken over decisions. Being linear rather than quadratic in fragmentation, and conditioned on the mask, it asks two mask sources comparably hard questions, and splitting and merging appear separately as low recall and low precision. Blind spot: a fragment pair the mask never brought within reach counts against nobody, so it cannot see a filament the mask lost entirely, and it is therefore always quoted with the number of decisions each mask posed.

Reference-instance recovery is the fraction of reference filaments recovered whole by one predicted instance. The dominant predicted instance of a filament is the one owning most of its fragments; coverage is that count over the filament's fragment count, purity the same count over every fragment that instance owns anywhere. A filament counts as recovered when the dominance is *mutual* and coverage and purity both reach 0.8. Purity is what stops a method that merges everything from scoring 1.0. It weights every filament equally, long or short, complementing the pairwise score's quadratic weighting. Blind spot: the two thresholds, which make it insensitive to how badly a filament that fails them was reconstructed.

Completeness, correctness and quality characterise the input *mask*: no reconstruction is opened to compute them and no method can change them. Both the reference axis and the mask under test are skeletonised and each tested against the other dilated by the tolerance. Completeness is tolerant recall of the reference axis, correctness tolerant precision of the test axis, quality the tolerant Jaccard index $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$ with TP the smaller of the two matched counts, so quality cannot exceed either component. Quality is also GraFT's published Jaccard index in its intended reading. Blind spot: the tolerance carries the measure, since at zero tolerance it reports raster placement rather than geometry, which is why it is never quoted without one.

Shared-pixel precision and recall are precision and recall over the pixels the reference gives to more than one instance, computed on the raw instance masks. They exist because the pairwise measure discards junction pixels and cannot see whether a crossing pixel was marked as jointly owned at all. Only a method emitting overlapping layers can score above zero on the recall; one emitting a hard partition scores exactly zero by construction, which is the point of measuring it. Blind spot: neither component distinguishes a pixel shared for the right reason from one shared because a chain was painted with a whole junction cluster, which is the over-marking Section 3.4 reports.

The runtime scaling exponent is the slope of an ordinary least-squares fit of log seconds on log foreground pixels, so an exponent b means cost grows as the b-th power of the amount of ink. It is reported instead of absolute time because these are independently optimised codebases, where absolute time measures implementation as much as algorithm. Blind spot: it describes a median trend and says nothing about the tail, so it is quoted beside the per-density medians and the censored-scene count; censoring biases it downwards for any method whose slowest scenes did not finish, which is why GraFT's is a lower bound.

Two measures were computed and are not endpoints. GraFT's *filament matched coverage* asks whether each reference filament was matched one-to-one to some detection, and PLECTA ranks *below* both comparators on it: at the densest stratum of GraFT's own generator, 0.798 against 0.998 for GraFT and 0.900 for DNAi. It is not an endpoint here because it counts existence rather than correctness: a method emitting more objects than the scene contains has a spare for every filament, and across the ladder each method's matched coverage is the smaller of one and its emission ratio to within 0.16, and also requiring a detection to cover half the axis it claims restores the order. *Centreline Dice* [42] was tried as an alternative matching criterion for detection F_1 and not adopted: it tests each side's skeleton against the other's *volume*, and a centreline grouping has none, so it degenerates into the placement test the tolerance exists to correct. No clDice result is reported in this paper.

B Comparator configuration

Section 3.3 states each adaptation and its cost; this is the mechanism behind it, for a reader checking that the comparison was run fairly. **DNAi** clusters junction pixels by single linkage and erases a disk at each cluster. At its released 10 px linkage distance, crossings closer than that chain into one cluster, which at 60 % coverage reaches 89 px across and is replaced by a single small erasure disk, leaving a whole tangle as one instance, which is why its shipped default is not a fair measure of it. Reducing the distance to 2 px on development scenes raised development F_1 and cut variation-of-information merge by about a factor of three, confirming the mechanism; the tuned configuration is the one reported. Two structural choices account for the residual difference: it commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can, and it attempts no junction above degree four, resolving 0.016 of the crossings above it. Those are about a tenth of the crossings, but they carry 37.2 % of same-filament fragment pairs on the degraded masks, one such junction severing every pair that spans it. **GraFT**'s errors are divisions rather than merges (variation-of-information split 1.166 against merge 0.392), which follows from a greedy longest-first path cover that cannot revise an early commitment once the scene collapses into one connected graph, as it does above 30 % coverage.